

- 6 The table shows the number of bottles of AndClean hand sanitizers, sold in three shops over the months of December 2019, January and February 2020.

	Selling Price (\$)	December 2019	January 2020	February 2020
Shop A	3.80	280	320	345
Shop B	3.50	250	265	280
Shop C	4.00	190	235	290

The selling price of AndClean hand sanitizers in each of the three shops can be represented by the matrix $\mathbf{P} = \begin{pmatrix} 3.80 & 3.50 & 4.00 \end{pmatrix}$.

- (a) Write down a 3×3 matrix $\mathbf{Q}$ to represent the number of bottles of AndClean hand sanitizers sold in each of the three shops over the three months.

Answer $\mathbf{Q} =$ _____ [1]

- (b) (i) State a 3×1 matrix $\mathbf{R}$ such that the product $\mathbf{QR}$ represents the total number of AndClean hand sanitizers sold in each shop for the three months.

Answer $\mathbf{R} =$ _____ [1]

- (ii) Evaluate the product $\mathbf{QR}$.

Answer $\mathbf{QR} =$ _____ [1]

- (c) Given that the cost price of a bottle of AndClean hand sanitizers is \$2.60, show that the profit each shop makes for selling a bottle of AndClean is given by $\mathbf{S} = \begin{pmatrix} 1.20 & 0.90 & 1.40 \end{pmatrix}$. [1]

- (d) (i) Evaluate the matrix $\mathbf{T} = \mathbf{S}(\mathbf{QR})$.

Answer $\mathbf{T} =$ _____ [1]

- (ii) Explain what the elements in $\mathbf{T}$ represent.

Answer _____

 _____ [1]

- (e) The World Health Organisation classified Covid-19 as a pandemic in March 2020. The number of bottles of AndClean hand sanitizers sold in March 2020 increased by 40% across all 3 shops as compared to February 2020.

Using scalar multiplication, find the number of bottles of AndClean sold in each of the three shops in March 2020.

Answer _____ [2]

- 7 The variables x and y are connected by the equation $y = \frac{1}{5}x^3 - \frac{4}{5}x^2$.

Some corresponding values of x and y are given in the following table.

x	-2	-1	0	1	2	3	4	5
y	-4.8	-1	0	-0.6	p	-1.8	0	5

- (a) Calculate the value of p .

Answer $p =$ _____ [1]

- (b) Using a scale of 2 cm to 1 unit, draw a horizontal x -axis for $-2 \leq x \leq 5$.
 Using a scale of 2 cm to 1 unit, draw a vertical y -axis for $-5 \leq y \leq 5$.
 On the grid provided, plot the points given in the table and join them with a smooth curve. [3]